

Group	ID	Age	Education	CS/EE Background	Familiar Platform	Embedded Experience
Non-Expert	N1	18–24	Bachelor’s	No	NA	Never
	N2	18–24	Bachelor’s	No	NA	Never
	N3	18–24	Bachelor’s	No	NA	Never
	N4	18–24	Bachelor’s	No	Mijia	Never
	N5	25–34	Graduate	No	Apple HomeKit	Never
Expert	E1	18–24	Graduate	Yes	NA	Never
	E2	25–34	Graduate	Yes	Mijia	More than 3 years
	E3	18–24	Graduate	Yes	Mijia	Less than 1 year
	E4	25–34	Graduate	Yes	Home Assistant	More than 3 years
	E5	25–34	Graduate	Yes	NA	2 years
	E6	18–24	Graduate	Yes	NA	Never
	E7	18–24	Bachelor’s	Yes	NA	Less than 1 year
	E8	18–24	Bachelor’s	Yes	NA	Never
	E9	25–34	Graduate	Yes	NA	Never
	E10	18–24	Graduate	Yes	Apple HomeKit, Tuya, Mijia	2 years

Table 5: Demographic characteristics of the participants in user study.

variety of hardware every day, and writing code would require flipping through numerous documents. AutoBridge could potentially allow hardware engineers to work on multiple tasks simultaneously, exponentially increasing their work efficiency.” (E7)

Superiority Over Alternative Methods. During Task A1 and B1, expert-group participants were allowed to use any AI tools or online resources. However, in Task A1, only two participants with prior embedded development experience (E4 and E10) successfully completed the task, and both attributed their success to their background rather than to AI assistance. “Thanks to my EE and CS background—and my deep familiarity with the end-to-end workflow—I can hand-craft the integration code for an obscure device. I even tried using Claude directly, but its hallucinations were so bad that I was better off writing the code myself. Coding by hand without any automated feedback felt painful—I’d have to deploy just to discover what went wrong, then slog through endless debug cycles.” (E4) Other participants lacked relevant experience and quickly encountered the limitations of general-purpose AI tools. Hallucinations were common when trying to generate code in this domain-specific knowledge-intensive task, which may explain their failure to complete the task. “It’s not something that can be easily solved by just feeding it to AI; the amount of documentation involved is too vast, and it’s impossible to upload all of it for AI to reference.” (E7)

Differentiated User Demands. By comparing the SUS scores across groups, we observed that Non-Expert participants gave higher ratings for both overall SU and usability than the Expert group. This may be because Non-Experts did not participate in the manual coding tasks (Task A1 and B1) and were less concerned with technical implementation details. They generally prioritized ease of use. “The whole workflow is crystal-clear and asks almost nothing of me, so the experience feels slick and fast.” (N5) In contrast, Expert participants who have experienced the difficulty of manual integration, had a deeper appreciation for the technical barriers and thus evaluated AutoBridge based on its improvements in efficiency and accuracy for experienced developers. Many drew comparisons to their professional workflows and expressed curiosity about how AutoBridge achieves its functionality. “Writing integration code for a device isn’t too hard for me—though at the start I do spend time learning the platform’s coding standards, which can be a bit time-consuming. I really like AutoBridge’s iterative debug approach: it spares users from complex analysis so even beginners can use it easily, lowering the barrier to customizing a smart home. On top of that, AutoBridge is pretty efficient—much faster than hand-coding—and perfect for hardware engineers looking to boost productivity.” (E10)

Potential Improvement. Participants also proposed suggestions for extending AutoBridge’s capabilities. One participant suggested that AutoBridge could output a breakdown of intermediate reasoning steps alongside code changes, similar to tools like Cursor, making it a valuable learning tool for developers. “If, in the future, AutoBridge could, like Cursor, output step-by-step analysis alongside each code change, it would be an absolute game-changer for professional developers.” (E4) One participant proposed integrating AutoBridge with a visual recognition system to enable device identification and response validation via camera input, eliminating the need for human feedback. “It would be great if AutoBridge could be connected to a visual recognition system, allowing it to automatically identify devices through a camera and check whether the device’s responses during debugging are correct. This could enable full automation of the entire process, which would be amazing.” (E2)

D Integration Code for Self-Developed Device in User Study

```
1 import logging
2 import requests
```